

MEDTECH VENTURE STUDIO

NEO NATAL JAUNDICE BLANKET

IBEX MEDICAL SOLUTION

PHASE II ASSESSMENT

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OVERVIEW



DISEASE STATE

Neonatal jaundice, prevalent in newborns due to high bilirubin levels.



PROCEDURE

Utilizing a phototherapy blanket to effectively reduce bilirubin levels.



PROBLEM

Wired phototherapy devices restrict newborn and caregiver mobility, limiting essential mother-child interaction during treatment.



OVERVIEW

NUMBERS



DISEASE STATE

Neonatal jaundice is a common condition, affecting approximately 60% of newborns—equivalent to nearly 2.5 million infants annually in the U.S.



PROCEDURE

Of these, around 800,000 newborns require hospitalization due to jaundice-related complications.



PROBLEM

Roughly 220,000 infants undergo phototherapy as a treatment for jaundice.



PROBLEM CONFIRMATION INTERVIEW

SARAH JANE SPRANGLER- NICU NURSE, PRIMARY CHILDREN'S HOSPITAL

Prevalence & Treatment

- Common in newborns (NICU & non-NICU)
- Treated with phototherapy (Bili lights & blankets)
- Effective in days, but blankets less efficient

Economic & Access Issues

- Most NICU babies on Medicaid, affecting affordability
- Severe cases hospitalized; mild cases sent home
- Rural families struggle to access treatment

Challenges

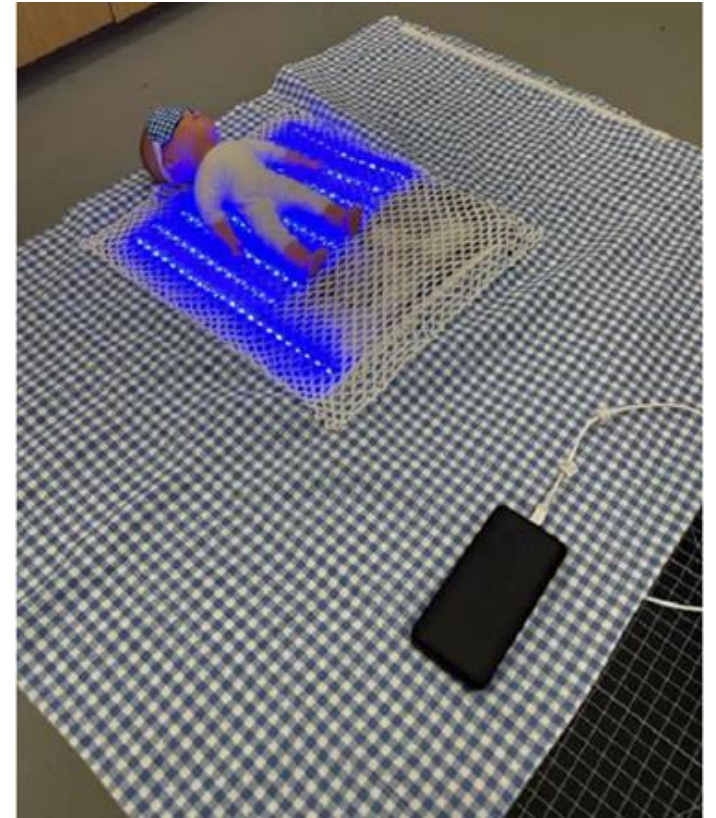
- Bili blankets too small for larger babies
- Hard to swaddle, causing discomfort
- Limited light exposure (only baby's back)
- Need for wireless options to improve mobility

Proposed Improvements

- Design larger, adaptable Bili blankets
- Use Velcro straps for better usability
- Develop wireless phototherapy for flexibility
- Address economic & rural access barriers

PROPOSED SOLUTION

- **Swaddle-based phototherapy device** with portable, washable, and water-resistant materials.
- Detachable **light pad** with 608 COB LEDs, reflective and photo-diffusive sheets for even light.
- Safety-focused with **temperature sensor** that auto-shuts off at unsafe heat levels.
- Operates using **Qi induction charging**, no direct wiring in swaddle—maximizes safety.



VALUE PROPOSITION

Accessibility

Portable design makes treatment accessible at home, decreasing hospital stays.

Safety and Comfort

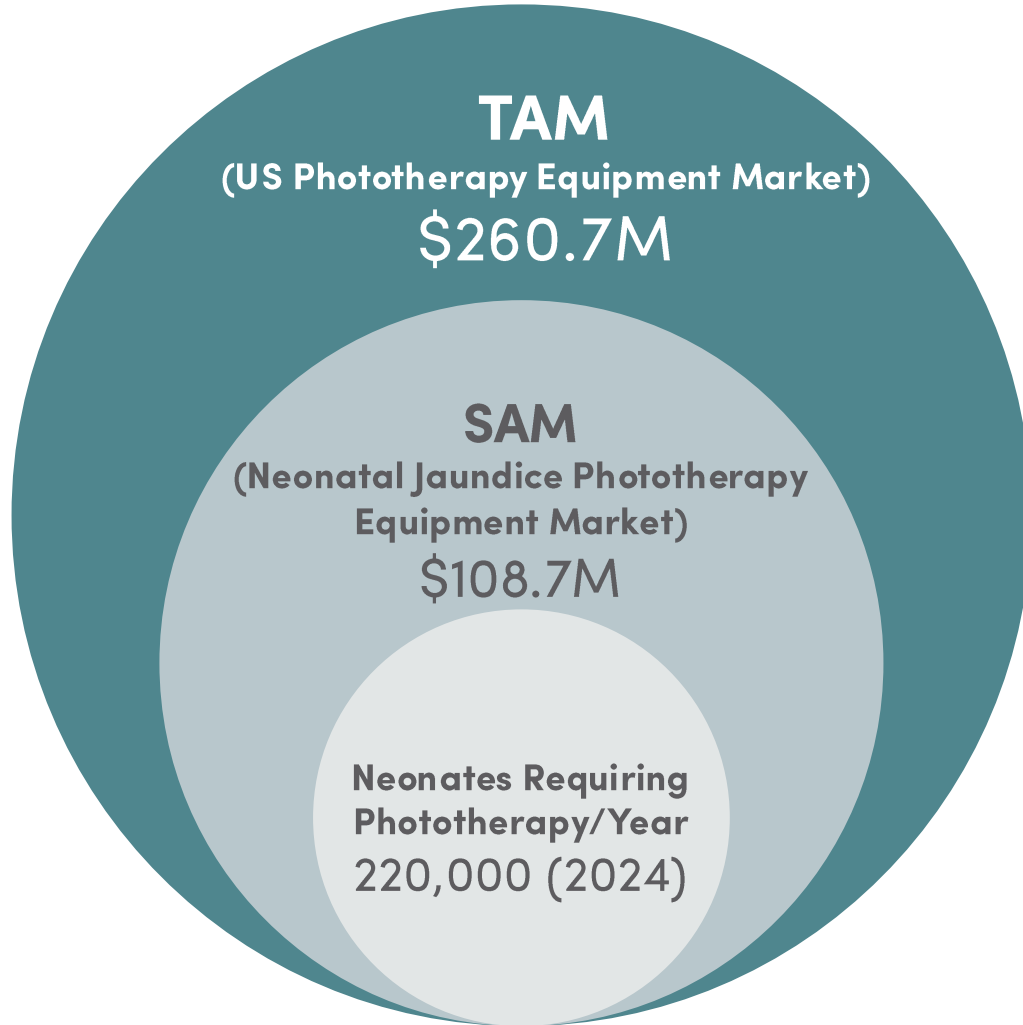
Includes overheating protection, promotes caregiver interaction with skin-to-skin contact.

Cost Effectiveness

More affordable and convenient than traditional phototherapy units.



MARKET OVERVIEW



The U.S. holds **43% of the global neonatal phototherapy market**, growing at **4.7% CAGR**.

Home-use segment is the fastest-growing in the U.S., with a **10% CAGR**, driven by demand for convenient, accessible care.

Rise in **LED and fiber-optic** use is shaping next-gen phototherapy devices.

Actual need may be higher due to **limited access** to devices—many are advised to use sunlight instead.

As **affordable, portable** phototherapy options grow, treatment rates are expected to increase significantly.

CURRENT SOLUTIONS



MOTIF MEDICAL

BILI-TOUCH BLANKET **\$3,000**

A portable LED blanket for neonatal jaundice, using 450–460 nm blue light. Battery-powered and cable-free, it features a temperature sensor and LCD display for safety



GE HEALTHCARE

BILISOFT **\$5,000**

A hospital-grade phototherapy system with a fiber-optic pad, delivering 450–460 nm blue light. It offers large coverage, soft disposable covers, and supports kangaroo care, perfect for NICU but available for home use



NATUS MEDICAL

NEO BLUE **\$3000**

A hospital-grade phototherapy device with 450–475 nm blue LED light. Lightweight and quiet, it fits cribs or incubators, supports double therapy, and enables bonding during treatment in clinical or home settings.

REGULATORY ASSESSMENT



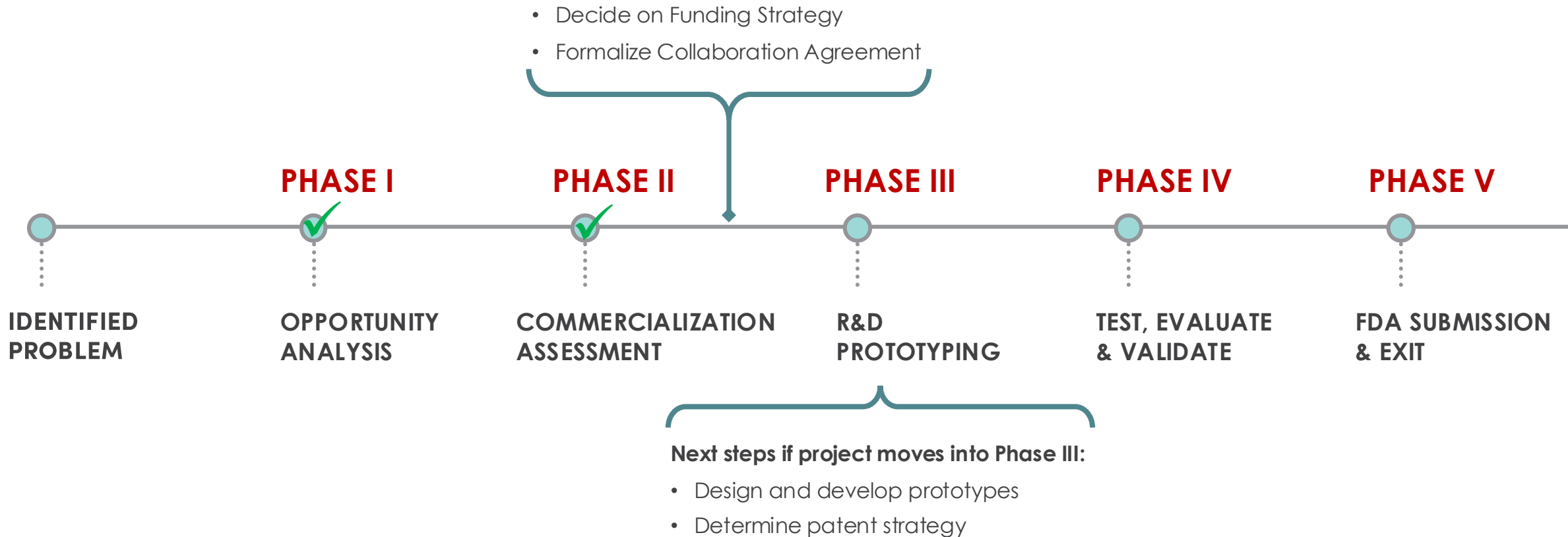
INTENDED SUBMISSION: Traditional 510(k)

CLASSIFICATION INFORMATION					
FDA CFR No.	PRODUCT CODE	CLASS	REVIEW PANEL	PRODUCT CODE DESCRIPTION	SUBMISSION TYPE
880.5700	LBI	II	General Hospital	Neonatal Phototherapy Unit	510(k)
	PDH			Neonatal Phototherapy Blanket	

REIMBURSEMENT & COVERAGE

CODING			
CODE	DESCRIPTION	RENTAL EQUIPMENT FLOOR	RENTAL EQUIPMENT CEILING
E0202	Phototherapy (bilirubin) light with photometer	72.63	85.45
S9098	Home visit, phototherapy services(e.g., Bili-lite), including equipment rental, nursing services, blood draw, supplies, and other services,	Not payable by Medicare	

NEXT STEPS





MEDTECH VENTURE STUDIO

THANK YOU

innovation for a better tomorrow



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